MySQL Assignment 2 -Querying Data

Overview:

The focus of this assignment is on data insertion, querying, filtering, aggregation, and result interpretation while ensuring consistency with the existing database design.

Database: Employee

Tables:

● Table1: Department

Attributes:

○ Department_id – Primary Key

○ Department_name

● Table2: Location

Attributes:

○ Location_id – Primary Key

○ Location_name

● Table3: Employees

Attributes:

○ Employee_id – Primary Key

○ Employee_name

○ Gender

○ Age

○ Hire_date

○ Designation

○ Salary

○ Department_id – Foreign Key referencing Department (Department_id)

○ Location_id – Foreign Key referencing Location (Location_id)

Tasks

Clause & Operators:

1. DISTINCT VALUES:

- A query to retrieve distinct salaries from the Employees table.

```
-- 1.DISTINCT VALUES
SELECT DISTINCT Salary FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

And the Distinct Salary Count

```
SELECT COUNT(DISTINCT Salary) as Distinct_Salary_Count FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Distinct_Salary_Count
11

2. ALIAS (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
-- 2.ALIAS (AS)
SELECT Age Employee_Age,Salary Employee_Salary FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00

3. WHERE CLAUSE & OPERATORS:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
-- 3.WHERE CLAUSE & OPERATORS
SELECT * FROM Employees WHERE Salary > 50000 and Hire_Date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_ID	Location_ID
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	12	1
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT * FROM Employees WHERE Designation IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_ID	Location_ID
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
UPDATE Employees SET Designation ='Data Scientist' WHERE Employee_ID = 5004;
SELECT * FROM Employees WHERE Employee_ID = '5004';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_ID	Location_ID
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	70000.00	8	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Sorting and Grouping Data:

1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

```
-- 1.ORDER BY
SELECT * FROM Employees ORDER BY Department_ID, Salary DESC;
```

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_ID	Location_ID
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2	1
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	75000.00	3	2
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

```
-- 2. LIMIT
SELECT * FROM Employees WHERE YEAR(Hire_Date)='2018' ORDER BY Hire_Date LIMIT 5;
```

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_ID	Location_ID
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

```
-- 3. AGGREGATE FUNCTIONS
SELECT SUM(E.Salary) Total_Fianance_Salary FROM Employees E
JOIN Departments D
ON E.Department_ID = D.Department_ID
WHERE D.Department_Name = 'Finance';
```

Total_Fianance_Salary
170000.00

- Find the minimum age among all employees.

```
-- Minimum Age
SELECT MIN(Age) Minimum_Age FROM Employees;
```

Minimum_Age
25

4. GROUP BY:

- List the maximum salary for each location.

```
-- 4. GROUP BY
SELECT L.Location, MAX(E.Salary) Maximum_Salary FROM Employees E
JOIN Location L
ON E.Location_ID = L.Location_ID
GROUP BY L.Location;
```

Location	Maximum_Salary
Bangalore	95000.00
Chennai	95000.00
Hyderabad	90000.00
Pune	100000.00

- Calculate the average salary for each designation containing the word 'Analyst'.

```
SELECT Designation,AVG(Salary) Average_Salary FROM Employees
Where Designation LIKE '%analyst%'
GROUP BY Designation;
```

Designation	Average_Salary
Data Analyst	60000.0000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

- Find departments with less than 3 employees.

```
-- 5.HAVING
SELECT E.Department_ID,D.Department_Name, COUNT(*) Employee_Count
FROM Employees E
JOIN Departments D
ON E.Department_ID = D.Department_ID
GROUP BY Department_ID
Having COUNT(*) < 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_ID	Department_Name	Employee_Count
1	Software Development	1
5	Product Management	2
6	Content Creation	2
7	Finance	2
9	Research and Development	1
11	Business Development	2
13	Operations	1

- Find locations with female employees whose average age is below 30.

```
SELECT L.Location, AVG(E.Age) Average_Age
FROM Employees E
JOIN Location L
ON E.Location_ID = L.Location_ID
WHERE Gender = 'F'
GROUP BY L.Location
HAVING AVG(E.Age) < 30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

Location	Average_Age
Bangalore	27.3333
Chennai	26.7500
Pune	29.2000

Joins:

1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

```
-- 1.INNER JOIN
SELECT E.Employee_Name, E.Designation, D.Department_Name
From Employees E
INNER join Departments D
ON E.Department_ID = D. Department_ID;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Name	Designation	Department_Name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science

2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

```
-- 2.LEFT JOIN
SELECT D.Department_Name, COUNT(E.Employee_ID) Total_Employees
FROM Departments D
LEFT JOIN Employees E
ON D.Department_ID = E.Department_ID
GROUP BY Department_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_Name	Total_Employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1
Software Development	1

3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location.

If no employees are assigned to a location, display NULL for employee name.

```
-- 3. RIGHT JOIN
SELECT L.Location, E.Employee_Name
FROM Employees E
RIGHT JOIN Location L
ON E.Location_ID = L.Location_ID;
```

Result Grid | Filter Rows: | Export:

Location	Employee_Name
Bangalore	Aaradhya Iyer
Bangalore	Dhruv Shetty
Bangalore	Myra Verma
Bangalore	Vihaan Mohan
Bangalore	Kabir Nair
Bangalore	Anika Paul
Bangalore	Vivaan Singh
Bangalore	Arjun Kumar
Chennai	Revansh Singh

4. CROSS JOIN:

- Show all possible combinations of departments and locations.

```
-- 4.CROSS JOIN
SELECT D.Department_Name, L.Location
FROM Departments D
CROSS JOIN Location L;
```

Result Grid | Filter Rows: | Export:

Department_Name	Location
Business Development	Pune
Business Development	Hyderabad
Business Development	Chennai
Business Development	Bangalore
Content Creation	Pune
Content Creation	Hyderabad
Content Creation	Chennai
Content Creation	Bangalore
Customer Support	Pune

Result 48

5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

```
-- 5.SELF JOIN
SELECT E1.Employee_Name Employee_1,E2.Employee_Name Employee_2, E1.Department_ID
FROM Employees E1
JOIN Employees E2
ON E1.Department_ID = E2.Department_ID
AND E1.Employee_ID < E2.Employee_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_1	Employee_2	Department_ID
Vihaan Singh	Avni Iyengar	3
Vihaan Singh	Vivaan Singh	3
Reyansh Singh	Atharv Yadav	12
Reyansh Singh	Kabir Nair	12
Reyansh Singh	Arjun Kumar	12
Aaradhya Iyer	Arnav Rao	10
Aaradhya Iyer	Vihaan Mohan	10
Kiara Malhotra	Dhruv Shetty	8
Kiara Malhotra	Diya Jha	8
Anvi Chaudhary	Kiaan Desai	11
Dhruv Shetty	Diya Jha	8

Windows function:

- Write a window function query to rank employees by salary using rank().

```
-- Windows function
-- RANK_()
SELECT Employee_Name, Salary,
RANK() OVER (ORDER BY Salary DESC) Salary_Rank
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

Employee_Name	Salary	Salary_Rank
Zoya Khan	100000.00	1
Avni Iyengar	100000.00	1
Myra Verma	95000.00	3
Aryan Shetty	95000.00	3
Anushka Singh	90000.00	5
Aarav Nair	90000.00	5
Sara Iyer	90000.00	5
Ishaan Kumar	85000.00	8
Advik Kapoor	85000.00	8
Reyansh Singh	80000.00	10

- Write a window function query to rank employees by salary within each department using DENSE_RANK()

```
-- DENSE_RANK()
SELECT Employee_Name,Department_ID, Salary,
DENSE_RANK() OVER (PARTITION BY Department_ID ORDER BY Salary DESC) Salary_Rank_By_Dept
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Name	Department_ID	Salary	Salary_Rank_By_Dept
Aarav Nair	1	90000.00	1
Anushka Singh	2	90000.00	1
Anika Paul	2	70000.00	2
Saanvi Patel	2	60000.00	3
Avni Iyengar	3	100000.00	1
Vivaan Singh	3	75000.00	2
Vihaan Singh	3	60000.00	3
Zoya Khan	4	100000.00	1
Aadhya Iyengar	4	60000.00	2
Ishika Patel	4	55000.00	3
Aryan Shetty	5	95000.00	1

- Write a window function query, Running total salary by department

```
-- RUNNING TOTAL
SELECT Employee_Name, Department_ID, Salary,
SUM(Salary) OVER (PARTITION BY Department_ID ORDER BY Salary) Running_Total_Salary
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Name	Department_ID	Salary	Running_Total_Salary
Aarav Nair	1	90000.00	90000.00
Saanvi Patel	2	60000.00	60000.00
Anika Paul	2	70000.00	130000.00
Anushka Singh	2	90000.00	220000.00
Vihaan Singh	3	60000.00	60000.00
Vivaan Singh	3	75000.00	135000.00
Avni Iyengar	3	100000.00	235000.00
Ishika Patel	4	55000.00	55000.00
Aadhya Iyengar	4	60000.00	115000.00
Zoya Khan	4	100000.00	215000.00
Sara Iyer	5	90000.00	90000.00

Conclusion:

From this Assignment, learnt about Querying, filtering, aggregation, and result interpretation .